



ENERGETIKA
ENERGETICS

Published from 2004
Ministry of Press and Information
of Azerbaijan Republic,
Registration number 3337, 07.03.2011

ISSN 1816-2126
Number 01, 2025
Section: English

ECOENERGETICS

HONORARY EDITOR IN CHIEF: Fagan G. Aliyev
SENIOR EDITOR: Rashad G. Abaszade
MANAGING EDITOR: Elmira A. Khanmamadova

INTERNATIONAL REVIEW BOARD

Arif Pashayev, Azerbaijan	Rafiq Aliyev, Azerbaijan	Turhan Vaziroglu, USA
Vagif Abbasov, Azerbaijan	Fuad Hajizadeh, Azerbaijan	Shiro Takada, Japan
Vagif Farzaliyev, Azerbaijan	İsmayil Aliyev, Azerbaijan	Luca Di Palma, İtalia
Natig Cavadov, Azerbaijan	Nazim İmanov, Azerbaijan	Yuriy Tabunshikov, Russian
Khadiyya Khalilova, Azerbaijan	Ali Guliyev, Azerbaijan	Mithat Kaya, Turkey
Farhad Aliyev, Azerbaijan	Leyla Mammadova, Azerbaijan	Elvin Aliyev, UK
Sakin Cabarov, Azerbaijan	Nazim Shamilov, Azerbaijan	Emre Gür, Turkey
Adil Azizov, Azerbaijan	Salahaddin Yusifov, Azerbaijan	Volodymyr Kotsyubynsky, Ukraine
Azer Mammadov, Azerbaijan	Mazahir İsayev, Azerbaijan	Matlab Mirzayev, Russian
Nurmammad Mammadov, Azerbaijan	Yusif Aliyev, Azerbaijan	Olga Kapush, Ukraine
Akif Alizadeh, Azerbaijan	Adil Abdullayev, Azerbaijan	Aitbek Aimukhanov, Kazakhstan
Rahim Alakbarov, Azerbaijan	Alakper Hasanov, Azerbaijan	Maksym Stetsenko, China
Gurban Eyyubov, Azerbaijan	Sevinj Malikova, Azerbaijan	Aida Bakirova, Kyrgyzstan
Samad Yusifov, Azerbaijan	Tarlan Huseynov, Azerbaijan	Bakhtyar Soltabayev, Kazakhstan
Ruslan Nuriyev, Azerbaijan	Tamella Naibova, Azerbaijan	Dzmitry Yakimchuk, Belarusiya
Agali Guliyev, Azerbaijan	Oktay Salamov, Azerbaijan	Saifullah Khalid, İndia
Tural Nagiyev, Azerbaijan	Asif Pashayev, Azerbaijan	Anitha Velu, İndia
Latif Aliyev, Azerbaijan	İlman Hasanov, Azerbaijan	Koushik Guha, India
		Nitesh Dutt, India
		Neeraj Kumar Bhoi, India
		Akbar Haghi, Portugal
		Muhammad Sultan, Pakistan

TECHNICAL EDITORIAL BOARD

SENIOR SECRETARY: İmran Y. Bayramov, Afig M. Nabiyev, Karim G. Karimov, Turan A. Nahmatova, Nigar V. Abbasova, Rashid Y. Safarov, Nuranə A. Zohrabbayli, Seynura A. Hasanova, Kamila A. Cafarli.

PUBLISHING OFFICE

5, M.Rahim, AZ-1073, Baku Azerbaijan

Tel.: 99412 538-23-70, 99412 538-40-25 Fax: 99412 538-51-22

E-mail: info@ieeacademy.org ekoenergetics@innovationresearch.az

Internet: <http://ieeacademy.org> www.innovationresearch.az

CONTENTS

1	Advances and Trends in Micro- and Nano-Fabrication Rashad Abaszade, Elmira Khanmamadova	1-23
2	Application of drones to hybrid energy systems: operating cost reduction and advantages N.T.Vahabli	24-28
3	Economic and environmental studies in the sustainable development of renewable energy in Karabakh M.Huseynova	29-32
4	Comparative analysis of oil-contaminated soil remediation methods A.Azizov, S.Huseynli	33-35
5	Development of an Automated Control System for Crude Oil Heater and Its U.Rzayev, İ.Bayramov	36-38
6	Electromagnetic waves and their effects on the human organism T. Huseynov, B. Davudov	39-42
7	Investigating the impact of climate change on the caspian sea level and ecological conditions A.Khalilova, N.Farzaliyeva	43-48
8	Laser strengthening of CP65+5%MF abrasive material for gears of oil pumps A.Guliyev, Y.Jahangirova	49-56
9	Plastic Alternatives: Bioplastics and Nanotechnology L.H. Memmedova, F.F. Ismayilova	57-59
10	Rehabilitation, modernization and expansion of electrical networks in liberated territories K.Gulmammadov	60-68
11	Renewable energy sources in the territory of east zangezur and Karabakh S.Gasimova, G.Mammadova	69-71
12	Sustainable urbanization and green infrastructure for reducing carbon emissions S.Gasimova, N.Əhədzadə	72-75
13	Solutions for optimizing heat dissipation on the printed circuit board of the drive circuit of a DC motor Jamil Gunashov, Rashad Abaszade	76-82
14	Prompt-based forecasting of monthly solar irradiation in the fuzuli district using ai models Mobil Maharramov	83-86
15	Industrial waste processing and utilization: a study of Sumgayit technopark C.A.Gurbanov	87-90
16	Artificial Intelligence and Geology F.M.Hajizadeh, N.Y.Ahadi	91-95
17	Energy Sector Vulnerability Fidan Abdullazade, Farhad Aliyev	96-99
18	Water Degradation Challenges in Azerbaijan's Karabakh Region S. R. Yusifova	100-103
19	Waste collection and utilization at sumgait technologies park C.A.Gurbanov	104-106

Energy Sector Vulnerability

Fidan Abdullazade

Azerbaijan University of Architecture and Construction, Baku, Azerbaijan

fidan.turkzade@gmail.com

Abstract. The impact of climate change on the energy sector is of particular importance for industrialized and energy-centered regions such as the Absheron Peninsula. This area contains the main oil and gas production and processing infrastructure of Azerbaijan. Climate change, in particular, increasing temperatures, intensification of extreme weather events and depletion of water resources, negatively affects the sustainability of energy infrastructure. The sensitivity of technological equipment used in energy production to heat and humidity, as well as overloading of electrical networks, further increase these risks. In addition, climate change requires a restructuring of energy policy against the background of the need to transition to renewable energy sources. The development of alternative sources such as solar and wind energy can not only create resilience to climate risks, but also play an important role in the process of decarbonizing the economy. However, this transition process requires a complex and coordinated approach. In this regard, risk-based planning, development of sustainable infrastructure and technological modernization in energy production are key priorities in order to reduce the vulnerability of the energy sector to climate change. At the same time, maintaining the balance between energy security and climate stability requires a strategic approach at the regional and national levels.

Keywords: Climate change, energy security, renewable energy, green transition, diversification, greenhouse gas emissions, fossil fuel consumption, energy efficiency

Corresponding author, email: fidan.turkzade@gmail.com

1.Introduction

Climate change is one of the most serious challenges of our time, having a profound impact not only on ecological systems, but also on the sustainability of economic sectors. One of the areas where this impact is felt most acutely is the energy sector. Global climate change — in particular, increasing temperatures, disruption of precipitation patterns, intensification of extreme weather events, and rising sea levels — is creating significant pressures on energy production, transmission, and consumption. Energy infrastructure in the Absheron Peninsula, one of Azerbaijan's strategic industrial regions, is of particular importance for the country's economic development and energy security. However, the region's ecological vulnerability and vulnerability to climate change pose additional risks to the sustainable operation of energy systems. The vulnerability of the energy sector to climate change is closely linked to both the physical availability of energy resources and technological and institutional capabilities. The exploitation and processing of energy sources such as oil and gas can face challenges in terms of productivity and safety as a result of high temperatures and dusty weather conditions. At the same time, damage and weakening of energy infrastructure (power plants, transmission lines, pipelines, etc.) due to extreme weather events can lead to interruptions in energy supply. All these factors create critical risks in terms of energy security, economic stability and social well-being. The vulnerability of the energy sector in the Absheron Peninsula to climate change will be analyzed, the main risk factors will be identified and possible adaptation strategies to increase the resilience of the sector will be considered[3].

2.Analysis

The impact of climate change on the energy sector has been a widely discussed issue at the global and regional levels in recent years. In a region as strategically important as the Absheron Peninsula in terms of both energy production and consumption, these impacts are even more significant. The region's climatic conditions, increasing heat waves, decreasing precipitation, and extreme weather events pose a serious threat to the sustainability of energy infrastructure. The Absheron Peninsula is a major energy hub for Azerbaijan, and the infrastructure located there —

including oil and gas fields, refineries, power plants, and distribution networks — is vulnerable to climate change to varying degrees. Rising air temperatures increase the demand for cooling systems, which results in a sharp increase in electricity demand during the summer months. At the same time, the decline in water resources used in energy production (especially for cooling purposes at thermal power plants) may jeopardize the stability of energy production. On the other hand, rising sea levels and the risk of coastal erosion pose additional challenges for Absheron's marine infrastructure. Oil and gas production platforms and terminals need more protection against sea level changes and stormy weather conditions. This requires additional technical measures and investments. Against the background of global energy transition processes, the region's oil and gas-based economic structure is considered more vulnerable to climate change. Thus, international commitments to reduce carbon emissions and trends in the transition to renewable energy sources call into question the long-term sustainability of existing energy models in Absheron. Therefore, in order to increase the sustainability of the energy sector, it is necessary to develop alternative energy sources (solar, wind, etc.), strengthen energy efficiency measures and adapt existing infrastructure to climate change[12]. Thus, climate change affects the energy sector on the Absheron Peninsula in several ways: the risk of damage to physical infrastructure, instability in the balance of energy production and consumption, limited water resources and the need for economic adaptation. This situation requires adapting energy policy to new realities and applying proactive approaches to managing climate risks.

Table 1. Climate Change Risks, Impacts, and Adaptation Measures in the Energy Sector of the Absheron Peninsula

Climate Change Risks	Impacts on the Energy Sector	Potential Adaptation and Response Measures
Rising temperatures	Increased demand for cooling; peak electricity consumption	Implementation of energy efficiency measures; deployment of smart energy management systems
Decreased precipitation, drought	Water shortages affecting thermoelectric power plant operations	Adoption of alternative cooling technologies (e.g., air-cooled systems); improved water resource management
Rising sea levels	Damage risk to coastal energy infrastructure such as terminals and platforms	Climate-proofing of marine facilities; construction of coastal protection barriers
Extreme weather events (storms, strong winds)	Physical damage to transmission lines, power stations, and offshore installations	Reinforcement of infrastructure; development of emergency backup and redundancy systems
Shifts in global energy policy	Economic vulnerability due to dependency on oil and gas markets	Diversification toward renewable energy sources; structural transition within the energy sector
Urbanization and growing energy demand	Overloading of the grid; risk of energy shortages	Deployment of smart grids; digital solutions for demand-side energy management

The table 1 presents a structured overview of key climate change risks affecting the energy sector in the Absheron Peninsula, their specific impacts, and possible adaptation strategies. It highlights how rising temperatures and decreasing precipitation lead to increased energy demand and reduced water availability for power generation, while extreme weather events and sea-level rise pose direct threats to infrastructure stability. Furthermore, the table emphasizes the implications of global energy policy shifts on a region heavily dependent on fossil fuels, as well as the growing pressure on the electricity grid due to rapid urbanization. The adaptation measures proposed include technological upgrades, energy diversification, smart grid deployment, and strategic infrastructure reinforcement — all aimed at enhancing the sector's climate resilience and long-term sustainability [5]. The energy sector in the Absheron Peninsula is increasingly exposed to the adverse impacts of climate change, both in terms of its physical infrastructure and its structural dependence on fossil fuels. Rising temperatures, more frequent heatwaves, reduced precipitation, and extreme weather events threaten the reliability and efficiency of energy production and distribution systems. These risks are particularly significant in a region like Absheron, where urban energy demand is high and infrastructure is often aged or insufficiently adapted to new climatic conditions. Higher ambient temperatures decrease the efficiency of thermoelectric power generation, while increased cooling demand during the summer months puts significant stress on the electricity grid. The likelihood of system overloads and blackouts rises accordingly. In addition, the reduced availability of water resources, which are critical for cooling systems in conventional power plants, can disrupt the continuity of energy supply. Offshore infrastructure, such as oil and gas extraction platforms in the Caspian Sea, also faces heightened vulnerability. Sea level rise, coastal erosion, and stronger storms pose direct threats to these installations, increasing the risk of operational failures, environmental incidents, and economic losses. These factors highlight the urgency of strengthening resilience and integrating climate risk assessments into energy planning and investment decisions. Given the Absheron Peninsula's continued reliance on hydrocarbon-based energy, the region is also at risk from global economic and policy shifts favoring decarbonization. As the world moves toward cleaner energy sources, regions heavily invested in oil and gas must adapt or face economic marginalization. Therefore, energy diversification through the expansion of renewable sources such as solar and wind power is not only environmentally necessary but also economically strategic. To mitigate these vulnerabilities, it is essential to modernize energy infrastructure, invest in smart grid technologies, and promote energy efficiency. Additionally,

institutional reforms and strategic planning must prioritize climate adaptation and integrate sustainability into long-term energy policy. Without these proactive measures, the energy sector in the Absheron Peninsula will remain exposed to increasing climate-related disruptions with broad implications for national energy security and socio-economic development [14]. This growing vulnerability of the energy sector also underlines the need for improved governance and cross-sectoral coordination. Effective climate adaptation in the energy domain cannot be addressed in isolation; it requires collaboration with urban planning, water resource management, environmental protection, and disaster risk reduction bodies. Only through integrated planning can the complex interdependencies between energy infrastructure and climate systems be managed effectively. Capacity building and knowledge transfer are equally critical components. Local energy providers, policymakers, and technical specialists must be equipped with up-to-date information on climate risks, technological innovations, and best practices in climate-resilient energy planning. This includes fostering partnerships with international organizations, research institutions, and private sector actors engaged in sustainable energy solutions. Public awareness and stakeholder engagement are also key to achieving a resilient energy future. Communities in the Absheron Peninsula, particularly those in vulnerable urban and coastal zones, must be included in planning and decision-making processes. Encouraging behavioral change, such as energy conservation practices, alongside institutional reforms, can create a bottom-up push toward sustainability. Financing mechanisms should be developed to support adaptation efforts. These may include climate-resilient infrastructure investments funded through public-private partnerships, green bonds, or international climate finance instruments. Azerbaijan's potential access to global climate funds and sustainable development financing frameworks can serve as a catalyst for transforming its energy sector in line with climate adaptation priorities[1-14].

3.Conclusion

The table presents a structured overview of key climate change risks affecting the energy sector in the Absheron Peninsula, their specific impacts, and possible adaptation strategies. It highlights how rising temperatures and decreasing precipitation lead to increased energy demand and reduced water availability for power generation, while extreme weather events and sea-level rise pose direct threats to infrastructure stability. Furthermore, the table emphasizes the implications of global energy policy shifts on a region heavily dependent on fossil fuels, as well as the growing pressure on the electricity grid due to rapid urbanization. The adaptation measures proposed include technological upgrades, energy diversification, smart grid deployment, and strategic infrastructure reinforcement — all aimed at enhancing the sector's climate resilience and long-term sustainability. These observations demonstrate that the energy sector in the Absheron Peninsula is at a critical juncture where inaction could lead to compounding vulnerabilities, while timely and strategic interventions could turn challenges into opportunities for sustainable transformation. The intersection of environmental pressures, infrastructural constraints, and economic dependency on hydrocarbons makes it essential to adopt a forward-looking approach that integrates climate risk management into national energy planning. By prioritizing renewable energy development, investing in resilient and smart infrastructure, and aligning energy policies with international climate goals, Azerbaijan can significantly reduce systemic risks and foster greater energy security. Moreover, enhanced institutional coordination, transparency in energy governance, and active engagement with global climate finance mechanisms will further support the country's adaptation efforts. Addressing the climate vulnerability of the energy sector in the Absheron Peninsula is not only an environmental imperative but also a socio-economic necessity. Through comprehensive policy reforms and practical implementation of adaptation strategies, it is possible to ensure the continuity, reliability, and sustainability of energy services in the face of growing climate challenges.

References

1. Asian Development Bank. (2021). *Climate Risk Country Profile: Azerbaijan*. 24 pages. [<https://www.adb.org/sites/default/files/publication/707466/climate-risk-country-profile-azerbaijan.pdf>] [adb.org](https://www.adb.org)
2. World Bank. (2023). *Country Climate and Development Report for Azerbaijan*. 112 pages. [<https://www.worldbank.org/en/country/azerbaijan/publication/country-climate-and-development-report-for-azerbaijan>] [worldbank.org](https://www.worldbank.org)
3. United Nations Framework Convention on Climate Change (UNFCCC). (2024). *Initial National Adaptation Plan: Azerbaijan*. 86 pages. [https://unfccc.int/sites/default/files/resource/2024_NAP_Azerbaijan.pdf] unfccc.int
4. Red Cross Red Crescent Climate Centre. (2024). *Azerbaijan Climate Country Profile*. 20 pages. [https://www.climatecentre.org/wp-content/uploads/RCCC-Country-profiles-Azerbaijan_2024_final.pdf] [climatecentre.org](https://www.climatecentre.org)
5. ReliefWeb. (2024). *Azerbaijan: Climate Country Profile (November 2024)*. 16 pages. [<https://reliefweb.int/report/azerbaijan/azerbaijan-climate-country-profile-november-2024>] reliefweb.int+3Le Monde+3
6. Rzaeva, S., & Tagiyev, A. (2024). Assessment of the potential for use of thermal waters of the Absheron Peninsula as an alternative energy source. *Reliability: Theory & Applications*, 19(SI), 511–519. 9 pages.

F.Abdullazade, Ecoenergetics, vol. 30, № 1, pp.96-99, (2025)

- [https://gnedenko.net/Journal/2024/SI_062024/RTA_SI062024-062_06_Sevinj%20Rzaeva%20Allahverdi%20Tagiyev.pdf]gnedenko.net
7. Organization for Security and Co-operation in Europe (OSCE). (2017). *Climate Change and Security in the South Caucasus*. 60 pages. [<https://www.osce.org/files/f/documents/4/9/331921.pdf>][osce.org](https://www.osce.org)
 8. Jafarov, E., & Hasanova, S. (2023). Green economic potential of Baku and Absheron-Khizi economic regions. *Polish Journal of Environmental Studies*, 32(1), 183–194. 12 pages. [<https://www.pjoes.com/pdf-183794-109481?filename=Green+Economic+Potential.pdf>][pjoes.com](https://www.pjoes.com)
 9. Mammadov, T. S., & Balapour, S. (2015). Climate change impacts on Azerbaijan biodiversity in the Caspian Sea. *Procedia Environmental Sciences*, 29, 123–124. 2 pages. [<https://doi.org/10.1016/j.proenv.2015.07.200>]
[Wikipedia](#)
 10. United Nations Development Programme (UNDP). (2015). *Climate Change and Disaster Risk Reduction Snapshot: Azerbaijan*. 4 pages. [<https://www.undp.org/sites/g/files/zskgke326/files/migration/eurasia/UNDP-RBEC-Azerbaijan-Climate-Change-and-Disaster-Risk-Reduction-Snapshot.pdf>][UNDP](https://www.undp.org)
 11. World Bank. (2013). *Building Resilience to Climate Change in South Caucasus Agriculture*. 120 pages. [<https://documents.worldbank.org/curated/en/193691468012673593/pdf/Building-resilience-to-climate-change-in-South-Caucasus-agriculture.pdf>][World Bank](https://documents.worldbank.org)
 12. International Bank of Azerbaijan. (2024). *Impacts and Consequences of Climate Change*. 30 pages. [https://abb-bank.az/storage/uploads/files/1730982201_abb-climate-report-eng-publish.pdf][ABB](https://abb-bank.az)
 13. European Union. (2024). *EU4Climate: Azerbaijan*. 2 pages. [<https://eu4climate.eu/azerbaijan/>][EU4Climate](https://eu4climate.eu)
 14. International Federation of Red Cross and Red Crescent Societies (IFRC). (2024). *Azerbaijan Climate Change Data: Emissions and Policies*. 10 pages. [<https://www.climatewatchdata.org/countries/AZE>]
[climatewatchdata.org](https://www.climatewatchdata.org)

A photograph of a wind farm with several white wind turbines on a green, hilly landscape under a cloudy sky. The image is framed by a dark green curved border at the top.

ECO

ENERGETIKA
ENERGETICS

A graphic of a water splash or wave, rendered in a light green and white color scheme, positioned on the right side of the lower half of the cover.